

Flexible Cable Simulation with Rigid Connectors in IsaacSim

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1 Introduction

This report presents a method for modeling and simulating a flexible cable connected to rigid connectors within NVIDIA IsaacSim. Since IsaacSim does not natively support deformable 1D cables, the cable is approximated using a chain of rigid capsule segments connected by constrained joints, providing realistic bending and torsional behavior.

2 Approach

The cable is modeled as a chain of **60 capsule-shaped rigid bodies** connected by **D6 joints** (6 degree-of-freedom joints). Each D6 joint locks all three translational axes and constrains the three rotational axes with cone and twist limits, producing cable-like flexibility.

Two rigid connectors are attached to the cable:

- **Top connector** (blue cube): A rigid block fixed to the world frame via a fixed joint. The cable's first link attaches to its bottom face through a D6 joint with limited rotation.
- **Bottom connector** (red cube): A weighted rigid block attached to the cable's free end via a fixed joint, representing a plug or end-fitting.

A **green obstacle** (fixed cube) is placed in the scene so the cable can drape over it and demonstrate realistic contact interaction.

3 Simulation Parameters

Table 1: Cable and simulation parameters.

Parameter	Symbol	Value
Number of links	N	60
Total cable length	L	1.0 m
Link radius	r	4.0 mm
Link height (cylinder)	h	8.67 mm
Segment spacing	Δs	16.67 mm
Total cable mass	m_{total}	1.0 kg
Per-link mass	m_i	16.67 g
Cone limit (swing)	θ_c	8.0°
Twist limit	θ_t	5.8°
Linear damping	d_l	0.2
Angular damping	d_a	1.0
Joint damping	d_j	0.05
Physics time step	Δt	1/240 s
Solver position iterations		64
Solver velocity iterations		8
Solver type		TGS
CCD		enabled

Table 2: Rigid connector parameters.

	Top connector	Bottom connector
Size (cube side)	6 cm	4 cm
Mass	5.0 kg	0.5 kg
Constraint	Fixed to world	Free (weighted)

4 Results

The simulation was run for 10 seconds. Key frames captured during the simulation are shown in Figure ???. The cable is given an initial lateral velocity of 1.5 m/s at the bottom connector to induce swinging motion toward the obstacle.

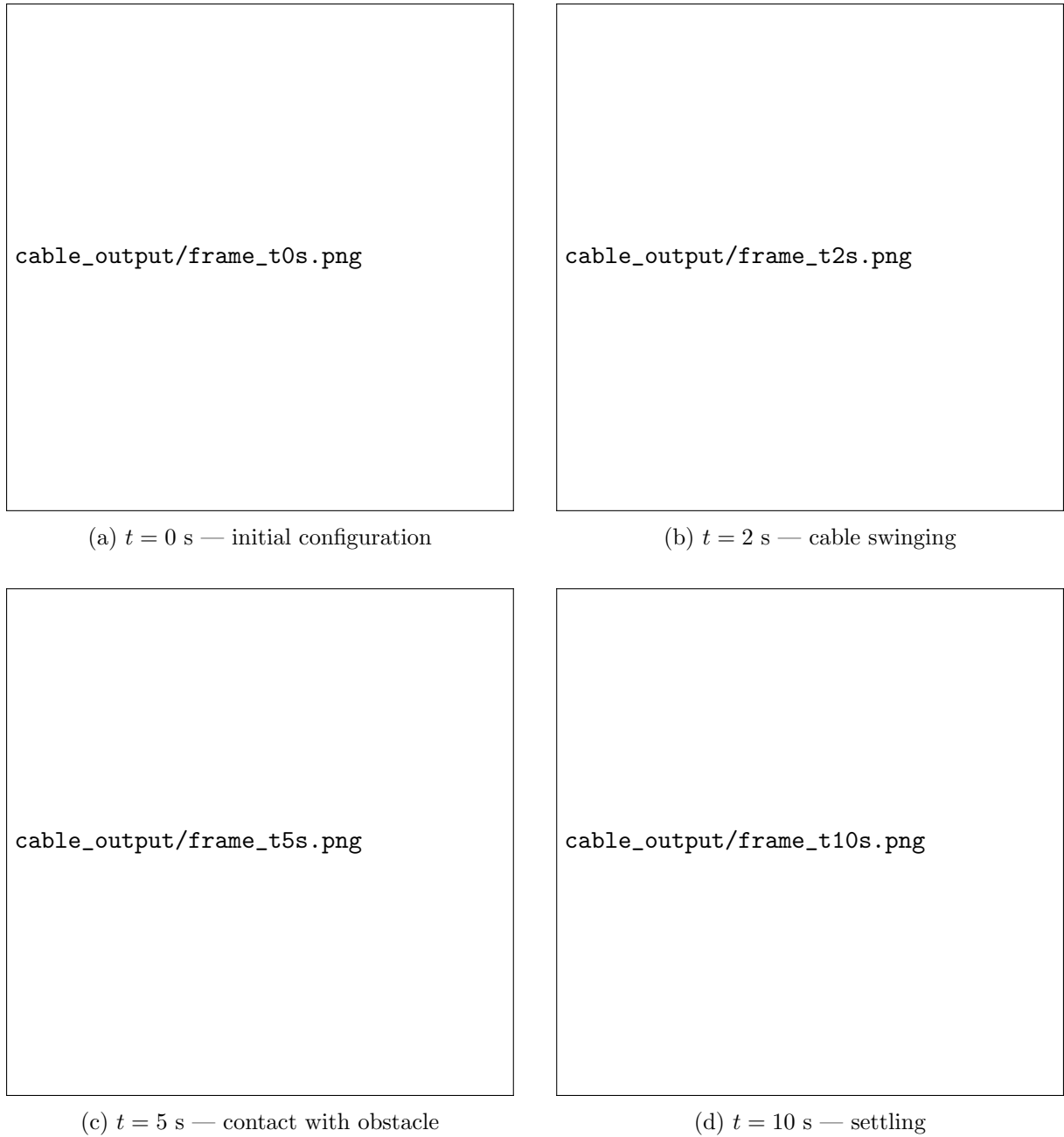


Figure 1: Key frames from the cable simulation.

A full 10-second video recording of the simulation is available in the supplementary file `cable.simulation.mp4`.

5 Challenges

- **Stability:** Long rigid-body chains are prone to jitter. High solver iteration counts (64 position / 8 velocity) and the TGS solver were required for stable behavior.
- **Thin-body collisions:** The small capsule radius (4 mm) required continuous collision detection (CCD) to prevent tunneling at high velocities.
- **Damping tuning:** Balancing linear, angular, and joint damping was critical — too little causes a frictionless pendulum chain; too much makes the cable overly rigid.

- **Joint limits:** The cone and twist limits control bending stiffness. Smaller angles produce a stiffer cable but can cause constraint violations if set too tight.

6 Conclusion and Next Steps

The capsule-chain approach successfully models a flexible cable connected to rigid connectors in IsaacSim, with realistic gravity, contact, and bending behavior. Possible next steps include:

- Integrating the cable with a robot manipulator for cable manipulation tasks.
- Exploring IsaacSim’s deformable body API as an alternative to the rigid-body chain approach.
- Adding friction and material properties for more realistic surface interactions.
- Parameterizing the cable for different lengths, stiffnesses, and diameters.